ANALYSIS OF DEEP NEURAL NETWORK LEARNING DYNAMICS THROUGH THE NTK LENS

Deeper or Wider ? A 1st answer under the NTK perspective

August 23, 2025

Janis AIAD

figures/umdlogo.jpg

CONTENTS

Abstract **4**

Executive summary **5**

Introduction **5**

Notations **6**

1 Preconditioning strategies for the NTK **9**
1.1 Multi-Component Multi-layer Neural Networks (MMNN) and Transformers 9
1.1.1 Feature Learning vs Kernel Regimes 9

2 MMNN Model Definition and Asymptotic Regime **10**
2.1 Personal contribution : NTK initialization for MMNN Architecture 10
2.2 Personal contribution : Parameterization and Training Regime 11
2.3 Personal contribution : Asymptotic Regime 12
2.4 Personal contribution : Signal Propagation and the Edge of Chaos (EOC) 13

3 Personal contribution: NTK formulas for MMNNs **13**
3.1 NTK for a Single-Layer MMNN 13
3.2 From Gradient Products to an Expectation 14
3.3 Integral Formulation and Parameter Mapping 14
3.4 Final Analytical Expression 14
3.5 Personal contribution : Recursion for Multi-Layer MMNNs 15
3.6 NNGP Kernel $\Sigma^{(\ell)}$ Recursion 15
3.7 Personal contribution: NTK Kernel $\Theta^{(\ell)}$ Recursion 15
3.7.1 Personal contribution : Parameter Influence on NTK Preconditioning 16
3.8 Personal contribution: Two Hidden Layer MMNN Architecture NTK analysis 16
3.8.1 Final Result 17

4 Personal contribution: Ongoing research and conclusion **19**
4.1 Conclusion: A New Paradigm for NTK Analysis 21

A Appendix **23**
A.1 Detailed approach for Transformers and Softmax Analysis 23
A.1.1 Problem Statement: Moments of the Softmax Function 23
A.1.2 Personal contribution currently in progress : Approximation Strategy: Saddle-Point Method 23
A.1.3 Personal contribution currently in progress : Cumulant Expansion Framework 24
A.1.4 High-Dimensional Analysis through Word Embedding Characterization 25
A.1.5 Correlated Gaussian Matrix Generation 25
A.1.6 Multidimensional Hermite Polynomials and Isserlis-Wick Theorem 26
A.2 Personal contribution: Detailed Calculations for Two Hidden Layer MMNN 28
A.2.1 Calculation of activation's correlations 28
A.2.2 Calculation of $\mathbf{I}_{11}$ (Probability) 28
A.2.3 Special Case: Zero Means ($\mu_1 = \mu_2 = 0$) 28
A.2.4 Proof of the Recursion for the NTK 28
A.3 Fisher and Kibble Distributions 31

B Deep Narrow Neural Networks **32**
B.1 Scaled NTK at Initialization 32

B.2 Research Perspectives for Deep Narrow Networks 32

ABSTRACT

This report presents the results of a research internship focused on studying the learning dynamics of deep neural networks. We explore the underlying mechanisms through the Neural Tangent Kernel (NTK) formalism [**jacot2018neural**], with particular attention to finite-width corrections [**seleznova2022ntk_beyond**]. Our work revolves around the decomposition of the NTK via Sobolev training, the analysis of its reproducing kernel Hilbert space (RKHS), and the study of extreme eigenvalues. We present theoretical bounds for the smallest eigenvalue of the NTK using random matrix theory (RMT) techniques. These results are then applied to analyze different architectures, including feed-forward neural networks (FCNN), matrix memory neural networks (MMNN), and transformers. We also discuss the implications of our findings for scientific machine learning (SciML) applications and propose directions for ongoing research. We voluntarily omit some results that are not yet ready to be published.

INTRODUCTION

The study of learning dynamics in deep neural networks constitutes a fundamental research area that has gained significant attention in recent years. Our approach focuses on the **desingularization of Sobolev training** by decomposing the problem into two distinct contributions: an independent Kand a Pcontribution. This decomposition relies on the commutation between integral operators and the fractional Laplace operator, which provides a powerful mathematical framework for understanding neural network behavior. The detailed calculations of these contributions are presented in Appendix A.2.

Our analysis continues with the study of Kpreconditioning through a freeze weights method, illustrated by neural networks factorizing low-rank weight matrices. This approach is supported by a rigorous approximation theorem that we develop. For classical feed-forward networks (FCNN), we develop a **finite-width correction** that proves particularly fruitful in understanding the gap between theory and practice. Finally, we explore Transformers as an alternative solution, offering a fundamentally new learning paradigm that challenges traditional assumptions about neural network training. A complete theoretical analysis of integrals with Gaussian measures and softmax functions for Transformers is detailed in Appendix A.1.

Across these different architectures, we study feature learning regimes and their scaling behavior with network width. Our results rely on statistical physics tools to characterize model convergence and asymptotic behavior, drawing connections between neural networks and well-established physical systems. This analysis enables a deeper understanding of finite-width corrections [**seleznova2022ntk_beyond**] and their practical implications for the optimization of deep neural networks in real-world applications.

NOTATIONS

Mathematical Notation Reference

Basic Mathematical Symbols

Symbol	Definition	Symbol	Definition
$\mathbb{R}$	Set of real numbers	$\mathbb{N}$	Set of natural numbers
$\mathbb{C}$	Set of complex numbers	$\mathbb{S}^{d-1}$	Unit sphere of dimension $d-1$
$\mathbb{E}[\cdot]$	Mathematical expectation	$\mathrm{Var}(\cdot)$	Variance of a random variable
$\mathrm{Cov}(\cdot,\cdot)$	Covariance between two variables	$\mathrm{Corr}(\cdot,\cdot)$	Correlation between two variables
$\nVdash\{\cdot\}$	Indicator function	dx	Differential element
$\Vert\cdot\Vert$	Euclidean norm	$\langle\cdot,\cdot\rangle$	Inner product
$\mathcal{O}(\cdot)$	Landau notation (big O)	$\tilde{\Omega}(\cdot)$	Landau notation (big Omega tilde)

Neural Network and Activation Functions

Symbol	Definition	Symbol	Definition
_/	activation function	$f(\mathbf{x};\boldsymbol{\theta})$	Neural network function
$\sigma_{(\cdot)}$	General standard deviation	σ^2	Activation variance parameter
L	Number of layers	n_k	Width of layer k

Neural Tangent Kernel (NTK)

Symbol	Definition	Symbol	Definition
$\mathbf{K}$	Discrete NTK matrix	K^∞	Infinite-width NTK operator
$K^L(\mathbf{x}_i,\mathbf{x}_j)$	NTK for L-layer network	$\mu_\ell^{(K)}$	NTK eigenvalues at degree ℓ
$\varrho(\rho)$	Correlation propagation function	$\rho_1(\mathbf{x},\mathbf{x}')$	Initial correlation

Mathematical Notation Reference (continued)

Sobolev Spaces and Training

Symbol	Definition	Symbol	Definition
$H^s(\Omega)$	Sobolev space of order s	$\mathbf{P}_s$	Discrete Sobolev operator matrix
P_s	Continuous Sobolev operator	$\mathcal{L}_{\text{Sobolev}}(f)$	Sobolev training loss function
D^k	k-th derivative operator	$\|f\|_{H^s}$	Sobolev norm of order s
$(I + (-\Delta)^{1/2})^s$	Fractional Laplacian operator	$\Delta_{\mathbb{S}^{d-1}}$	Laplace-Beltrami operator

Spherical Harmonics and Fourier Analysis

Symbol	Definition	Symbol	Definition
$Y_{\ell,p}$	Spherical harmonics	$N(d,\ell)$	Multiplicity of harmonics
$\hat{f}(\xi)$	Fourier transform of f	ξ	Frequency variable
$\sigma(x)$	Surface measure on sphere	ℓ	Degree index
p	Order index	∇	Gradient operator

Eigenvalues and Spectral Analysis

Symbol	Definition	Symbol	Definition
λ_ℓ	Eigenvalues at degree ℓ	$\lambda_{\min}(A)$	Minimum eigenvalue
$\lambda_{\max}(A)$	Maximum eigenvalue	$\mu_\ell^{(P_s)}$	Sobolev eigenvalues
$C(d,L)$	Constants depending on d, L	$\Gamma(\cdot)$	Gamma function
$P_\ell^{(\alpha)}(t)$	Gegenbauer polynomials	$\tilde{K}_L^\infty(t)$	Normalized NTK function

Matrix Operations and Training

Symbol	Definition	Symbol	Definition
A^T	Matrix transpose	$\|A\|_{op}$	Operator norm
$\|A\|_F$	Frobenius norm	$A \circ B$	Hadamard product
$\mathcal{T}_s$	Composite learning opera-	$a_{\ell,p}$	Sampling vectors

1
PRECONDITIONING STRATEGIES FOR THE NTK

1.1 Multi-Component Multi-layer Neural Networks (MMNN) and Transformers

We examine Multi-Component Multi-layer Neural Networks (MMNN) and Transformer architectures, focusing on NTK preconditioning techniques. This means we investigate how to change our architectural and training meta-parameters (activation function, residual connections, message passing).

At the time this report is written, most of the research is currently in progress.

For MMNN architectures, the kernel structure exhibits exhibits more stability. The preconditioning strategy decomposes the weight matrix into trainable/frozen weights, and make the loss landscape less dimensionnal, and hopefully more convex. We present all the mathematical results and proofs that are necessary to answer this question.

For Transformer models, the attention mechanism introduces non-linearities complicating kernel analysis. We introduce specialized techniques to handle softmax normalization within attention heads, in order to perform fast numerical experiments to compute intractable quantities appearing. We provide also physical approximations strategies that will be used to perform the final step, and an agenda to answer the question for this architecture.

1.1.1 • Feature Learning vs Kernel Regimes

The transition between feature learning and kernel regimes represents a fundamental dichotomy in deep learning theory. In the kernel regime, network parameters remain close to initialization and the model behaves as a linear predictor in the NTK-defined feature space. Conversely, the feature learning regime allows significant parameter evolution, enabling adaptation of internal representations.

The NTK spectrum, particularly its smallest eigenvalue $\lambda_{\min}$, determines convergence rates and optimization strategy. An ill-conditioned NTK suggests second-order methods are appropriate, while a well-conditioned NTK indicates first-order methods can achieve faster convergence. For MMNN architectures, different network components may operate in different regimes simultaneously, requiring careful analysis of heterogeneous optimization dynamics.

Conjecture from Pr Haizhao Yang: Global Minima Stability

For sufficiently wide MMNN and Transformer networks, global minima exhibit a high local stability propertiy ensuring convergence from neighborhoods around optimal solutions. The Hessian and NTK eigenvalues near global minima satisfy regularity conditions guaranteeing stable optimization dynamics.

This conjecture is supported by numerical experiments [Zhang et al, 2024] demonstrating super-convergence behavior in practical training scenarios. By means of local analysis Our goal now is to develop all the theoretical framework to have the tool to prove this conjecture.

2
MMNN MODEL DEFINITION AND ASYMPTOTIC REGIME

Objective and Scope

This work derives the NTK $\boldsymbol{\Theta}$ operator of a Multi-layer Mean-field Matrix-Normal Network **(MMNN)**. The kernel is computed with respect to a specific set of trainable parameters, under the standard **"NTK parameterization"** which ensures a well-defined infinite-width limit.

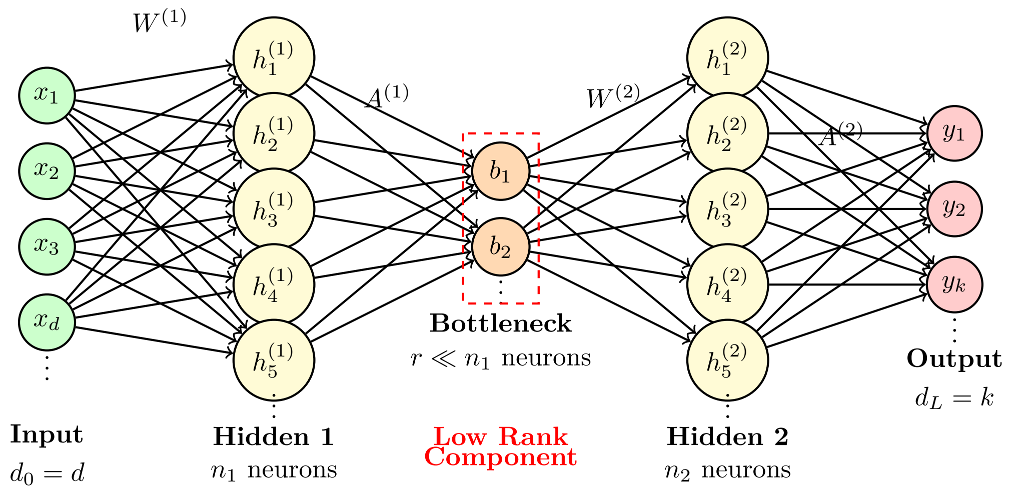


Figure 1: Architecture of a Multi-component Multi-layer Neural Network (MMNN) with a low-rank intermediate layer. The red dashed connections indicate the low-rank component that reduces dimensionality from n_1 to $r \ll n_1$ neurons. We only train the output weights $A^{(1)}$ and $A^{(2)}$.

2.1 Personal contribution : NTK initialization for MMNN Architecture

Definition: MMNN Architecture

We consider an **MMNN** of depth L. Let the input be $h^{(0)}(x) = x \in \mathbb{R}^{d_0}$. For each layer $\ell = 1, \dots, L$, the output $h^{(\ell)}$ is defined recursively:

$$h^{(\ell)}(x) = \frac{\sigma_A}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \,\text{\_/}\!\left(w_j^{(\ell)\top} h^{(\ell-1)}(x) + \beta b_j^{(\ell)} + \sigma_c c^{(\ell)}\right) \tag{1}$$

where n_ℓ is the width of layer ℓ and $h^{(\ell)} \in \mathbb{R}^{d_\ell}$.

Remark : Parameter Classification

The parameters for each layer are divided into two groups:
1. Trainable Parameters: The output weights $\{A_j^{(\ell)}\}_{j=1}^{n_\ell}$ and biases $c^{(\ell)}$ are optimized during training. The NTK **K** is computed with respect to these parameters.
Initialization:

- The weights $A_j^{(\ell)} \sim \mathcal{N}(0,1)$. The term $\frac{\sigma_A}{\sqrt{n_\ell}}$ is the **NTK parameterization**, ensuring kernel convergence as $n_\ell \to \infty$.
- The bias $c^{(\ell)} \sim \mathcal{N}(0,1)$ is initialized at zero with scaling σ_c.

2. Random Parameters: The input weights $w_j^{(\ell)}$ and biases $b_j^{(\ell)}$ are randomly initialized and **fixed** throughout training.

Insight : Architectural Philosophy

This random basis initialization is chosen on purpose, supported by approximation theorems [Zhang et al, 2024], with intuition from localizing high frequency variations. With this method, we reduce the number of parameters from n_ℓ^2 to $n_\ell * d_\ell$.
The NTK parameterization is chosen to ensure a well-defined infinite-width limit. The μ parameterization do not have this property.

2.2 Personal contribution : Parameterization and Training Regime

The choice of parameterization fundamentally determines the network's behavior in the infinite-width limit. We distinguish two main regimes that correspond to distinct learning philosophies. The first, NTK parameterization, favors stability and convergence toward a linear model, while the second, mean-field parameterization, promotes feature learning and non-linear dynamics.

Remark: Critical Parameterization Choice

The choice of scaling for the trainable parameters critically determines the network's behavior in the infinite-width limit.
1. NTK Parameterization (Our approach): We use the $\frac{1}{\sqrt{n_\ell}}$ scaling for output weights.

- Output function $h^{(\ell)}(x)$ maintains variance $\mathcal{O}(\sigma_A^2) \to$ **stable magnitude preservation**
- Gradient with respect to weights remains small: $\mathcal{O}(\frac{1}{\sqrt{n_\ell}}) \to$ **"lazy training" regime**
- Network dynamics converge to linear model behavior characterized by the Neural Tangent Kernel

2. Mean-Field (μ) Parameterization [Greg Yang et al., 2020]: Alternative $\frac{1}{n_\ell}$ scaling. In this parameterization, function outputs are of order $\mathcal{O}(\frac{1}{\sqrt{n_\ell}})$ while gradients remain of order $\mathcal{O}(1)$, enabling unit-scale parameter updates. This scaling is termed "mean-field" because each neuron's contribution is of order $\mathcal{O}(1)$, allowing collective influence of all neurons. This configuration promotes feature learning and non-linear dynamics.
Those initialization are opposite, and we will see that the NTK parameterization is the best for the MMNN architecture to have order one gaussianity in low rank layers.

Definition: Neural Network Gaussian Process (NNGP)

For a given layer ℓ, when $n_\ell \to \infty$, the output $h^{(\ell)}(x)$ converges in distribution to a composition of Gaussian processes whose covariance conditioned to the previous layer is determined by the NNGP kernel $\mathcal{K}_{NNGP}^{(\ell|\ell-1)}$.
Mathematically, we have:

$$\mathcal{K}_{NNGP}^{(\ell|\ell-1)}(x, x') = \text{Cov}\left[h^{(\ell)}(x), h^{(\ell)}(x') \mid h^{(\ell-1)}\right] \tag{2}$$

Crucial Remark: Composition of Gaussian Processes

Fundamental point: In this limit, the output of each layer $h^{(\ell)}$ converges in distribution to a **composition of Gaussian Processes (GP)** which is **not** itself a Gaussian process!
Consequently, the NNGP and NTK kernels for layers $\ell > 1$, which are functions of the output of the previous layer, remain **random quantities** (random fields). This is different from FCNNs but we use the same terminology on purpose. The recursive formulas describe the relationship between these random kernels.
This distinction is crucial because:

- For $\ell = 1$: $h^{(1)}$ is Gaussian $\to \mathcal{K}_{NNGP}^{(1)}$ and $\Theta^{(1)}$ are deterministic
- For $\ell > 1$: $h^{(\ell)}$ is composition of GP $\to \mathcal{K}_{NNGP}^{(\ell)}$ and $\Theta^{(\ell)}$ are random

The recursion captures the evolution of these kernel distributions through the layers.

2.3 Personal contribution : Asymptotic Regime

The entire NTK and NNGP formalism relies on the infinite-width limit giving gaussianity. Following the framework established by Jacot et al., we analyze the network in a specific asymptotic regime:

- The dimensions of the hidden layers, or widths, $n_1, n_2, \ldots, n_L$ are taken to infinity.
- The input and output dimensions of each layer $(d_0, d_1, \ldots, d_L)$ are kept fixed and finite.
- Critically, the limits are taken sequentially: first we let $n_1 \to \infty$, then $n_2 \to \infty$, and so on up to $n_L \to \infty$.
-

Remark: Sequential Limit as Key Theoretical Tool

This sequential limit is the key theoretical tool. When analyzing layer ℓ in the limit $n_\ell \to \infty$, the output of the previous layer, $h^{(\ell-1)}$, is a sum of an infinite number of i.i.d. random variables (by construction of layer $\ell - 1$). By the Central Limit Theorem, $h^{(\ell-1)}$ converges in distribution to a Gaussian Process. This allows us to replace the complex, random function $h^{(\ell-1)}$ with a GP whose covariance structure is precisely the NNGP kernel $\Sigma^{(\ell-1)}$. This is the foundation of the recursive calculation.

2.4 Personal contribution : Signal Propagation and the Edge of Chaos (EOC)

Before analyzing the kernel, we study the propagation of the signal's magnitude through the network at initialization. We are interested in the evolution of the variance of the layer outputs, defined as $q^{(\ell)} = \mathbb{E}[\|h^{(\ell)}\|^2]/d_\ell$.

For a ReLU activation and assuming centered parameters, the variance evolves according to the following recurrence relation (see Appendix for proof):

$$q^{(\ell)} = \sigma_c^2 + \frac{\sigma_A^2}{2}\left(q^{(\ell-1)}\mathrm{Tr}(\Sigma_w) + \beta^2\right) \tag{3}$$

This is an affine map $q^{(\ell)} = M \cdot q^{(\ell-1)} + C$. For the signal to propagate stably without vanishing ($M < 1$) or exploding ($M > 1$), the dynamics must be at the "Edge of Chaos", where the factor is exactly 1.

Edge of Chaos Condition

The condition for stable signal propagation is:

$$\frac{\sigma_A^2}{2}\mathrm{Tr}(\Sigma_w) = 1 \quad \Longrightarrow \quad \sigma_A\sqrt{\mathrm{Tr}(\Sigma_w)} = \sqrt{2} \tag{4}$$

We call the EOC initialization when adding the condition $\sigma_c^2 = 1$.

3

Personal contribution: NTK formulas for MMNNs

3.1 NTK for a Single-Layer MMNN

We begin by deriving the Neural Tangent Kernel for a single-layer MMNN. This result forms the base case for the multi-layer recursion. For simplicity, we consider a single scalar output, such that the network is defined as:

Single-Layer MMNN Definition

$$\boxed{h(x) = \frac{1}{\sqrt{n}}\sum_{j=1}^{n} A_j \,\mathrm{ReLU}(w_j^\top x + b_j) + c} \tag{5}$$

The trainable parameters are $\theta = \{A_j\}_{j=1}^{n} \cup \{c\}$. The parameters $\{w_j, b_j\}$ are fixed and randomly drawn, with $w_j \sim \mathcal{N}(0, \Sigma_w)$ and $b_j \sim \mathcal{N}(\mu_b, \sigma_b^2)$.

3.2 From Gradient Products to an Expectation

In the infinite-width limit ($n \to \infty$), the NTK converges to a rotation invariant kernel (hence zonal) as an expectation over the distribution of the random parameters (w, b). The full NTK is the sum of the kernel part from the weights $\{A_j\}$ and the part from the bias $\{c\}$.

NTK as Expectation

$$\boxed{\Theta^{(1)}(x, x') = \underbrace{\mathbb{E}_{w,b}\left[\_/(w^\top x + b)\ \_/(w^\top x' + b)\right]}_{\text{NNGP Kernel } \Sigma^{(1)}(x,x')} + \underbrace{\sigma_c^2}_{\text{from bias } c}} \tag{6}$$

3.3 Integral Formulation and Parameter Mapping

The NNGP kernel $\Sigma^{(1)}$ is an integral over the distributions of w and b:

$$\Sigma^{(1)}(x, x') = \int_{\mathbb{R}^d} \int_{\mathbb{R}} \_/(w^\top x + b)\ \_/(w^\top x' + b) p(w) p(b)\, db\, dw \tag{7}$$

A change of variables is performed from the parameter space (w, b) to the pre-activation space (g_1, g_2), where $g_1 = w^\top x + b$ and $g_2 = w^\top x' + b$. This transforms the high-dimensional integral into an expectation over a 2D bivariate Gaussian distribution. The parameters of this distribution are:

Remark: Bivariate Gaussian Parameters

- **Means:** $\mu_1 = \mathbb{E}[g_1] = \mathbb{E}[w^\top x + b] = \mu_b$. By symmetry, $\mu_2 = \mu_b$.
- **Variances:** $\sigma_1^2 = \mathrm{Var}(g_1) = \mathrm{Var}(w^\top x) + \mathrm{Var}(b) = x^\top \Sigma_w x + \sigma_b^2$. Similarly, $\sigma_2^2 = x'^\top \Sigma_w x' + \sigma_b^2$.
- **Covariance:** $\mathrm{Cov}(g_1, g_2) = \mathbb{E}[(w^\top x)(w^\top x')] + \mathrm{Var}(b) = x^\top \Sigma_w x' + \sigma_b^2$.

3.4 Final Analytical Expression

Theorem 3.1 (NNGP Kernel for a Single-Layer ReLU MMNN)**.** *Let the pre-activations (g_1, g_2) be jointly Gaussian with parameters (μ_1, σ_1^2) and (μ_2, σ_2^2) and correlation ρ. The NNGP kernel $\Sigma^{(1)} = \mathbb{E}[ReLU(g_1)ReLU(g_2)]$ is given by the decomposition:*

NNGP Kernel Decomposition

$$\boxed{\Sigma^{(1)} = \sigma_1\sigma_2 \cdot I_{12} + \mu_1\sigma_2 \cdot I_{01} + \mu_2\sigma_1 \cdot I_{10} + \mu_1\mu_2 \cdot I_{00}} \tag{8}$$

where the terms I_{ij} are moments of standardized bivariate normal variables (Z_1, Z_2) with thresholds $a_1 = -\mu_1/\sigma_1$ and $a_2 = -\mu_2/\sigma_2$.

3.5 Personal contribution : Recursion for Multi-Layer MMNNs

We now extend the single-layer result to deep networks. The recursion involves two distinct but intertwined random fields: the NNGP kernel $\Sigma^{(\ell)}$ and the NTK kernel $\Theta^{(\ell)}$.

Crucial Remark: Random Kernels in Deep Networks

An important fact is that the NNGP kernel $\Sigma^{(\ell)}$ is a random field that is NOT a gaussian process but a deep gaussian process, while the NTK kernel $\Theta^{(\ell)}$ is also a random field that is not a gaussian process. This is because the composition of a non-linear function with a Gaussian Process, $g(f(x))$, is not, in general, a Gaussian Process.

3.6 NNGP Kernel $\Sigma^{(\ell)}$ Recursion

The NNGP kernel describes the covariance of the network's output function at initialization. For any finite width, this output is a random function. In the infinite-width limit, it converges to a Gaussian Process. The base kernel $\Sigma^{(\ell)}$ is defined recursively.

NNGP Kernel Recursion

- **Base Case ($\ell = 1$):** The NNGP kernel for the first layer:

$$\boxed{\Sigma^{(1)}(x, x') = \mathbb{E}_{w,b}\left[\mathrm{ReLU}(w^\top x + b)\,\mathrm{ReLU}(w^\top x' + b)\right]} \tag{9}$$

- **Recursive Step ($\ell > 1$):** For subsequent layers:

$$\boxed{\Sigma^{(\ell)}(x, x') = \mathbb{E}_{w,b}\left[\mathrm{ReLU}\Big(w^\top h^{(\ell-1)}(x) + \beta b\Big)\,\mathrm{ReLU}\Big(w^\top h^{(\ell-1)}(x') + \beta b\Big)\right]} \tag{10}$$

3.7 Personal contribution: NTK Kernel $\Theta^{(\ell)}$ Recursion

The NTK is the Gram matrix of function gradients. For any finite width, it is a random matrix. The following recursion describes the relationship between these random kernels in the infinite-width limit.

NTK Recursion Formula

- **Initialization ($\ell = 0$):** $\Theta^{(0)} = 0$.
- **Recursive Step ($\ell \geq 1$):** The NTK for layer ℓ combines the NTK from previous layers with the contributions from the trainable parameters $A^{(\ell)}$ and $c^{(\ell)}$:

$$\boxed{\Theta^{(\ell)}(x, x') = \Theta^{(\ell-1)}(x, x') \cdot \left(\sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x')\right) + \sigma_A^2 \Sigma^{(\ell)}(x, x') + 1} \tag{11}$$

where $\sigma_c^2 = 1$ is the contribution from the trainable bias. $\Sigma^{(\ell)}$ is the base NNGP kernel defined above. $\dot{\Sigma}^{(\ell)}$ is the base derivative kernel:

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_{w,b}\left[\dot{\mathrm{ReLU}}\left(w^\top h^{(\ell-1)}(x) + \beta b\right) \dot{\mathrm{ReLU}}\left(w^\top h^{(\ell-1)}(x') + \beta b\right) w^\top w\right] \tag{12}$$

with $\dot{\mathrm{ReLU}}$ being the derivative of the activation function. Both $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random kernels for $\ell > 1$.

3.7.1 • Personal contribution : Parameter Influence on NTK Preconditioning

Remark: Activation and Bias Influence on NTK

The NTK behavior can be systematically controlled through two key parameters that serve as natural preconditioning mechanisms:
1. Activation Function Choice: The choice of σ directly and its convexity affect both $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ kernels. For ReLU, we obtain specific analytical forms that enable tractable computations, while alternative activations (sigmoid, tanh) modify the kernel's spectral properties and convergence behavior.
2. Bias Parameter β: The bias scaling β controls NTK preconditioning. Setting $\beta = 0$ yields centered pre-activations, while $\beta > 0$ improves training stability. This provides direct control over the kernel's spectral properties.

3.8 Personal contribution: Two Hidden Layer MMNN Architecture NTK analysis

Objective: Comparative Analysis of MMNN vs Standard MLP

The purpose of this analysis is to expose what the NTK should be when compared in the minimal setup against standard multi-layer perceptron. We consider a configuration with 2 hidden layers, where the low-rank structure is incorporated for MMNNs to demonstrate the fundamental differences in kernel behavior between these architectures.

The considered architecture is:

$$\begin{array}{ccccccccc} (x, y) & \xrightarrow{\sigma W_1} & h_1 & \xrightarrow{A_1 \cdot + c_1} & r & \xrightarrow{\sigma W_2} & h_2 & \xrightarrow{A_2 \cdot + c_2} & (x_2, y_2) \\ & & N & & (x_1, y_1) \text{ in } \mathbb{R}^r & & N & & \end{array}$$

Main Formula: Two Layer MMNN NTK

For $\beta = 0$, the **NTK** for two hidden layers is written as:

$$\mathbf{\Theta}^{(1)} = \mathbb{E}_w \left[\sigma(w^\top x)\sigma(w^\top x')\right] + 1 \tag{13}$$

$$\begin{aligned}\mathbf{\Theta}^{(2)} &= \mathbf{\Theta}^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_w \left[\dot{\sigma}(w^\top h^{(1)}(x))\dot{\sigma}(w^\top h^{(1)}(x'))w^\top w\right]\right) \\ &\quad + \mathbb{E}_w \left[\sigma(w^\top h^{(1)}(x))\sigma(w^\top h^{(1)}(x'))\right] + 1\end{aligned} \tag{14}$$

Ongoing Research: Deep Network Scaling via Lyapunov Analysis

The MMNN structure presented above is **particularly interesting** for analyzing deep networks with growing number of layers L. We can employ **Lyapunov analysis** to study the products of independent random variables that emerge through the layer-wise recursions. This approach allows us to derive the expected **scaling behavior with respect to depth** L, which is crucial for understanding the dynamics of very deep networks.

This scaling analysis through products of random matrices is precisely what we are pursuing in our **current investigations**, extending the two-layer results to arbitrary depth and characterizing the geometric properties of the resulting kernel evolution.

3.8.1 • Final Result

MAIN RESULT

For 2 inputs $x, y \in \mathbb{S}^{d-1}$ with cosine similarity ρ and a low-rank hidden layer r, the **NTK** for a two hidden layer EOC initialized MMNN is:

$$\boxed{\mathbf{\Theta}^{(2)}(x,y) = \tilde{\mathbf{\Theta}}^{(2)}(\rho) = \mathbf{\Theta}^{(1)}(\rho_1) \cdot \left(1 - \frac{\arccos(\rho_1)}{\pi}\right)\frac{2}{r} + \frac{2}{r}\mathbf{\Sigma}^{(1)}(\rho_1)\|x_1\|\|y_1\| + 1} \tag{15}$$

with $\mathbf{\Sigma}^{(1)}(\rho) = \frac{1}{\pi}\left(\sqrt{1-\rho^2} + \rho(1 - \arccos(\rho))\right)$

where the random variables follow:

- $\rho_1 \sim \text{Fisher}(\rho_1, r)$ is independent of the norms $(\|x_1\|, \|y_1\|)$
- $(\|x_1\|^2, \|y_1\|^2) \sim \text{Kibble}(r, \rho_1)$ are chi-squared variables with r degrees of freedom

High-Dimensional Behavior

For large values of r, Fisher's z-transformation gives $z = \text{arctanh}(\rho_1) \sim \mathcal{N}\left(\text{arctanh}(\rho) + \frac{\rho}{2(r-1)}, \frac{1}{r-3}\right)$.
The variance of ρ_1 decreases as $O(1/r)$. When $\rho \sim O(1/\sqrt{r})$, geometric noise masks the correlation.
For the normalized product $S/r = \|\mathbf{X}\|\|\mathbf{Y}\|/r$, we have:

$$\boxed{\frac{S}{r} \xrightarrow{r\to\infty} 1} \tag{16}$$

This convergence occurs with fluctuations of order $O(1/\sqrt{r})$.

Remark: Fast Decay of NTK Fluctuations

The NTK fluctuations decay rapidly with rate $O(1/r)$, which implies that we obtain **trivial eigenvalue bounds** provided we can compute the deterministic zonal function $\tilde{\mathbf{\Theta}}(\rho)$. This fast convergence rate ensures that the random components of the kernel become negligible for moderate values of r, allowing us to approximate the spectral properties using the deterministic part of the kernel. The theoretical challenge thus reduces to analyzing the spectrum of the limiting zonal kernel operator. Ongoing research is doing a taylor expansion of the NTK near $\rho = 1$ on that purpose.

Computational Insight: Linear Programming Formulation

Following Zhang et al.'s approach, uniform distributions for random weights $w_j^{(\ell)} \sim \text{Uniform}(\mathcal{S}^{d_{\ell-1}})$ and biases $b_j^{(\ell)} \sim \text{Uniform}[-1, 1]$ transform MMNN optimization into a **linear programming problem**. Since the random basis functions $\sigma(w_j^{(\ell)\top} h^{(\ell-1)}(x) + \beta b_j^{(\ell)})$ are fixed and the network output is linear in trainable weights $\{A_j^{(\ell)}\}$, the optimization becomes convex and exactly solvable. Another way to precondition the NTK is to choose a specific ad-hoc great initilization that we are trying to find in ongoing research.

4
PERSONAL CONTRIBUTION: ONGOING RESEARCH AND CONCLUSION

The initial results are promising and open several avenues for both numerical and theoretical investigation.

Numerical Investigations

- **Integration Accuracy:** Increase the network width/rank used for numerical integration to confirm that the observed phenomena are stable as we better approach the infinite-width limit.

Mathematical Investigations: Spectrum Analysis

A primary focus of future theoretical work will be to precisely characterize the spectrum of the MMNN's NTK operator using a trifecta of advanced analytical techniques.
Then we can focus on analyzing the influence of beta to ensure huge beta can or not precondition the NTK gram matrix.
The primary analytical approaches for the spectrum are:

- **Spectrum via Puiseux Series Expansion:** Following the work of Bietti and Bach [**bietti2021deepequalsshallowrelu**], we can analyze the Puiseux series expansion of the kernel function in the limits $\rho \to 1^-$ and $\rho \to -1^+$. The first non-integer exponent in this series expansion directly characterizes the asymptotic scaling of the NTK operator's eigenvalues.
- **Spectrum as a Zonal Operator:** A powerful approach is to analyze the NTK as a zonal kernel operator on the sphere. Following the methodology of Murray et al. [**murray2023characterizingspectrumntkpower**], we can seek a power series expansion of the kernel in terms of the Gram matrix ($K = x^\top x'$). This would allow for a direct characterization of its eigenvalues in terms of Gegenbauer polynomials.
- **Spectrum vs. Depth Analysis:** Adapt the proof methodology from Terjék and González-Sánchez [**terjek2025mlpseocspectrumntk**] to analyze how the NTK spectrum scales with depth L. This involves performing a power expansion of the kernel function with respect to the correlation ρ.

Mathematical Investigations: Extended Goals

Further mathematical goals include:

- **Connect to Function Decomposition:** Investigate using a uniform distribution for the random weights w. This choice may align better with the function approximation theory that originally motivated the MMNN architecture. The intuition is that pairs of weights with opposite signs (e.g., w and $-w$) could create basis functions that compensate for each other, effectively building localized basis functions over intervals defined by the pre-activations—a cornerstone of function decomposition.
- **Scaling Law Coefficient:** Derive the analytical form of the scaling law coefficient C_{scaling} from Experiment 2 and determine its dependency on hyperparameters like σ_A and β.
- **Rank-Magnitude Relationship:** Provide a rigorous theoretical explanation for the inverse relationship between layer rank and NTK magnitude, possibly leveraging Gaussian concentration of measure principles.
- **Constant Sine Kernel:** Investigate the analytical formulas for the Sine NTK to explain why it becomes nearly constant in our experimental setup, likely by analyzing the behavior of the correlation term ρ.
- **Theory of Evolving Kernels:** Develop a formal framework for analyzing the dynamics of the low-rank NTK, proving that it remains well-conditioned throughout training under certain assumptions.
- **Analysis of Trainable Activations (PSinTU):** A further avenue is to explore more expressive, trainable activation functions like Parametrized Sinusoidal Transform Unit (PSinTU). While this offers greater flexibility, the NTK analysis becomes significantly more complex as the basis functions themselves are no longer fixed. This will be tackled in a later phase of the research.

Conclusion: A New Paradigm for NTK Analysis

MMNN analysis through NTK theory reveals a fundamental trade-off and framework generalization. Standard NTK theory for MLPs requires quadratic overparameterization to guarantee static kernels and lazy training. MMNNs offer an alternative: entering analyzable, NTK-like regimes with linear parameter scaling.
The cost is kernel evolution: no longer deterministic and static, but a random field evolving during training. This randomness is manageable through high-dimensional probability tools. Deriving concentration bounds (Bernstein-type inequalities) on spectrum and kernel evolution provides robust, probabilistic optimization guarantees. This framework rigorously analyzes MMNN structural priors beyond idealized lazy training.

Outlook: PINNs Application and Wavelet Analogy

Major applications target Physics-Informed Neural Networks (PINNs) solving PDEs through neural network training satisfying governing equations. Since MMNNs derive from function decomposition theorems, their architecture naturally suits this purpose.
A powerful analogy connects MMNNs to wavelet transforms, particularly in high dimensions. Wavelets decompose functions onto fixed localized bases; MMNNs learn optimal combinations of rich, random function bases. Classical wavelets suffer dimensionality curses, becoming intractable for high-dimensional problems.
Networks perform automatic, adaptive directional domain decomposition. While neuron-partitioned regions grow exponentially, parameters scale linearly with width. This expressive power and parameter efficiency combination makes MMNNs powerful, tractable tools for high-dimensional PINN applications where classical wavelets fail.
Concrete progress requires network regularity. Many PDEs involve second-order derivatives (Laplacians). Applying rigorous optimization and approximation theorems requires twice-differentiable networks (C^2). Future directions include developing "Deep Smooth ReLU Network" (DSRN) MMNN versions using smoother activations, guaranteeing necessary regularity and unlocking broader theoretical tools for PINN analysis.

Remark: High-Dimensional Advantages

MMNNs overcome dimensionality curses through several mechanisms. In high dimensions, random vectors are nearly orthogonal, meaning correlation ρ between pre-activations concentrates around zero. The Funk-Hecke theorem analyzes kernel operators in this setting, showing ρ-dependent kernels have structured spectra. This ρ concentration implies NTK matrices become quasi-diagonal or take $(a-b)\mathbf{I}+b\mathbf{J}$ with $\mathbf{J}=\frac{1}{d}\mathbf{1}\mathbf{1}^\top$ the all ones matrix, simplifying spectral analysis along with standard deviation decay.

4.1 Conclusion: A New Paradigm for NTK Analysis

MMNN analysis through NTK theory reveals a fundamental trade-off and framework generalization. Standard NTK theory for MLPs requires quadratic overparameterization to guarantee static kernels and lazy training. MMNNs offer an alternative: entering analyzable, NTK-like regimes with linear parameter scaling.

The cost is kernel evolution: no longer deterministic and static, but a random field evolving during training. This randomness is manageable through high-dimensional probability tools. Deriving concentration bounds (Bernstein-type inequalities) on spectrum and kernel evolution provides robust, probabilistic optimization guarantees. This framework rigorously analyzes MMNN structural priors beyond idealized lazy training.

Major applications target Physics-Informed Neural Networks (PINNs) solving PDEs through neural network training satisfying governing equations. Since MMNNs derive from function decomposition theorems, their architecture naturally suits this purpose.

A powerful analogy connects MMNNs to wavelet transforms, particularly in high dimensions. Wavelets decompose functions onto fixed localized bases; MMNNs learn optimal combinations of rich, random function bases. Classical wavelets suffer dimensionality curses, becoming intractable for high-dimensional problems.

Networks perform automatic, adaptive directional domain decomposition. While neuron-partitioned regions grow exponentially, parameters scale linearly with width. This expressive power and parameter efficiency combination makes MMNNs powerful, tractable tools for high-dimensional PINN applications where classical wavelets fail.

Concrete progress requires network regularity. Many PDEs involve second-order derivatives (Laplacians). Applying rigorous optimization and approximation theorems requires twice-differentiable networks (C^2). Future directions include developing "Deep Suuper ReLU Network" [Yang et al., 2024] MMNN versions using smoother activations, guaranteeing necessary regularity and unlocking broader theoretical tools for PINN analysis.

A
APPENDIX

We focus here on understanding NTK for other architectures and handcomputations made through this comparison.

A.1 Detailed approach for Transformers and Softmax Analysis

We present here our approach to the analysis of the softmax function and the Gaussian measure integrals that appear in Transformers NTK analysis following the work of Hron et al.

Major Personal Contribution

This appendix presents the detailed calculations for the theoretical analysis of **Gaussian measure integrals** and **softmax functions**, specifically developed for **Transformer** models. These results constitute an original contribution to the mathematical understanding of **attention mechanisms**.

A.1.1 • Problem Statement: Moments of the Softmax Function

The concrete problem is to calculate the moments of the output of a softmax function applied row-wise to an $N \times P$ Gaussian matrix $\mathbf{X}$ with a correlated covariance structure. The softmax output $\mathbf{S}$ is given by:

$$\boxed{S_{ik} = \frac{e^{\mathbf{X}_{\mathbf{ik}}}}{Z_i}, \quad \text{where } Z_i = \sum_{l=1}^{P} e^{\mathbf{X}_{\mathbf{il}}}} \tag{17}$$

The objective is to calculate $\mathbb{E}[P(S_{ik})]$ over the distribution of $\mathbf{X}$. We note that the softmax structure is intrinsically linked to the logarithm of the partition function Z_i:

$$S_{ik} = \frac{\boldsymbol{\partial} \log(Z_i)}{\boldsymbol{\partial} X_{ik}} \tag{18}$$

Thus, $\mathbb{E}[P(S_{ik})] = \mathbb{E}[P(\frac{e^{\mathbf{X}_{\mathbf{ik}}}}{Z_i})] = \mathbb{E}[P(\frac{e^{\mathbf{X}_{\mathbf{ik}}}}{\sum_{l=1}^{P} e^{\mathbf{X}_{\mathbf{il}}}})]$ is the expectation of a polynomial of the derivatives of $\log Z_i$, making exact analytical calculation **intractable**.

A.1.2 • Personal contribution currently in progress : Approximation Strategy: Saddle-Point Method

We introduce a parameter β (inverse temperature) to obtain $S_{ik}(\beta) = e^{\beta X_{ik}} / \sum_l e^{\beta X_{il}}$. Following the asymptotic approach, we factorize the partition function Z_a:

$$\boxed{Z_a = \sum_{k=1}^{P} e^{\beta G_{ak}} = e^{\beta G_{a,\max}} \left(1 + \sum_{k \neq \max} e^{\beta(G_{ak} - G_{a,\max})}\right)} \tag{19}$$

We define $\delta_k = e^{\beta(G_{ak} - G_{a,\max})}$ for $k \neq \max$.

$$S_{ik}(\beta) = \frac{e^{\beta X_{ik}}}{\sum_{l=1}^{P} e^{\beta X_{il}}}, \quad \text{where } \beta = \frac{1}{T} \tag{20}$$

High Temperature Regime in Production Models

Current state-of-the-art transformer models operate predominantly in the **high temperature regime** with $T \in [3.3, \infty)$ corresponding to $\beta \in (0, 0.3]$ for non-creative applications. This regime is characterized by:

$$\boxed{T_{\text{typical}} \geq 3.3 \implies \beta_{\text{typical}} \leq 0.3} \tag{21}$$

In this high temperature setting, the softmax distribution approaches uniform behavior, leading to more diverse and less confident predictions. This contrasts with creative applications where lower temperatures ($T < 1$, $\beta > 1$) are used to produce more deterministic outputs.

A.1.3 • PERSONAL CONTRIBUTION CURRENTLY IN PROGRESS : CUMULANT EXPANSION FRAMEWORK

We employ the cumulant expansion approach, which is more physically motivated. First, we expand the logarithm:

$$\log(Z_a) = \log\left(e^{\beta G_{a,\max}}\right) + \log\left(1 + \sum_{k \neq \max} \delta_k\right) \tag{22}$$

$$\approx \beta G_{a,\max} + \left(\sum_{k \neq \max} \delta_k\right) - \frac{1}{2}\left(\sum_{k \neq \max} \delta_k\right)^2 + \dots \tag{23}$$

For the dominant output ($l = \max$):

$$S_{a,\max} = \exp\left(\beta G_{a,\max} - \log(Z_a)\right) \tag{24}$$

$$\approx \exp\left(-\sum_{k \neq \max} \delta_k + \frac{1}{2}\left(\sum_{k \neq \max} \delta_k\right)^2 - \dots\right) \tag{25}$$

$$\approx 1 - \sum_{k \neq \max} \delta_k + \left(\sum_{k \neq \max} \delta_k\right)^2 + \dots \tag{26}$$

For a perturbative output ($l \neq \max$):

$$\boxed{S_{al} = \exp(\beta G_{al} - \log(Z_a)) \approx \delta_l \left(1 - \sum_{k \neq \max} \delta_k + \dots\right)} \tag{27}$$

A.1.4 • High-Dimensional Analysis through Word Embedding Characterization

We analyze dot products between word embeddings in transformer architectures. Consider two independent word embedding vectors $\mathbf{e}_i$ and $\mathbf{e}_j$ from an embedding matrix $\mathbf{E} \in \mathbb{R}^{V \times d}$, where V is vocabulary size and d is embedding dimension.

Each component is initialized as i.i.d. $\mathcal{N}(0, \sigma_{\text{emb}}^2)$ with $\sigma_{\text{emb}} = 1/\sqrt{d}$ for unit variance. For two independent embeddings $\mathbf{e}_i, \mathbf{e}_j \in \mathbb{R}^d$, their dot product follows $Z = \mathbf{e}_i \cdot \mathbf{e}_j \sim \mathcal{N}(0, 1/d)$.

High-Dimensional Embedding Concentration

For the absolute value $|Z|$ following folded normal distribution with standard initialization:

$$\boxed{\mathbb{E}[|Z|] = \sqrt{\frac{2}{\pi d}}, \quad \text{Var}(|Z|) = \frac{0.36338}{d}} \tag{28}$$

The concentration property becomes pronounced as embedding dimension increases:

$$\boxed{\lim_{d\to\infty} \mathbb{E}[|Z|] = 0, \quad \lim_{d\to\infty} \text{Var}(|Z|) = 0} \tag{29}$$

Under isotropy assumption, embeddings are uniformly distributed on the unit sphere. For unit vectors $\mathbf{u}_i$ and $\mathbf{u}_j$, the dot product satisfies the cosine law:

$$\boxed{\mathbf{u}_i \cdot \mathbf{u}_j = \cos(\theta_{ij})} \tag{30}$$

where θ_{ij} is the angular distance. Under isotropy, the angle follows:

$$\boxed{\theta_{ij} \approx \frac{\pi}{2} + \frac{\mathbf{u}_i \cdot \mathbf{u}_j}{\sqrt{d}} + O(d^{-3/2})} \tag{31}$$

This concentration justifies that dot products between independent embeddings remain small and concentrated around zero as $d \to \infty$. This property ensures we can go through an expansion with small ρ

A.1.5 • Correlated Gaussian Matrix Generation

We adopt the generative framework from the analytical calculations using a global variable and row-specific components. Let $Z_0 \sim \mathcal{N}(0,1)$ be a global random variable. For each row $i \in \{1, \ldots, n\}$, let $Z_i = (Z_{i1}, Z_{i2}, \ldots, Z_{iP})$ be a vector of P independent standard normal variables.

The entries of the correlated Gaussian matrix $\mathbf{G}$ are constructed as:

$$\boxed{G_{ij} = Z_0 \cdot \rho_{ij} + Z_{ij}\sqrt{1 - \rho_{ij}^2}} \tag{32}$$

This construction ensures the desired covariance structure:

$$\text{Cov}(G_{ik}, G_{jl}) = \mathbb{E}[G_{ik}G_{jl}] = \rho_{ik}\rho_{jl} \tag{33}$$

The verification follows directly from the independence of the constituent random variables:

$$\mathbb{E}[G_{ik}G_{jl}] = \mathbb{E}[(Z_0\rho_{ik} + Z_{ik}\sqrt{1-\rho_{ik}^2})(Z_0\rho_{jl} + Z_{jl}\sqrt{1-\rho_{jl}^2})] \quad (34)$$
$$= \rho_{ik}\rho_{jl}\mathbb{E}[Z_0^2] + \delta_{i,j}\delta_{k,l}\sqrt{1-\rho_{ik}^2}\sqrt{1-\rho_{jl}^2}\mathbb{E}[Z_{ik}Z_{jl}] \quad (35)$$
$$= \rho_{ik}\rho_{jl} + \delta_{i,j}\delta_{k,l}(1-\rho_{ik}^2) \quad (36)$$
$$= \rho_{ik}\rho_{jl} \quad (37)$$

where we used $\rho_{ii} = 1$ for the diagonal terms. This generative model requires only $nP + 1$ independent random variables instead of the full n^2P variables needed for direct sampling.

Remark: Generative Model

This generative model requires $n^2 + 1$ independent random variables instead of the strictly necessary n^2P variables needed for direct sampling. We have slightly more variables than $n^2 - 1$, but we preserve a crucial property: the symmetry with respect to ρ in the definition, which is essential for integrating over a numerically coherent domain.

A.1.6 • Multidimensional Hermite Polynomials and Isserlis-Wick Theorem

We present our main tools to make this integration tractable at least numerically

Theorem A.1 (Multidimensional Hermite Polynomials and Orthogonality - Mehler (1866))**.** *For a d-dimensional Gaussian vector $\mathbf{X} = (X_1, \ldots, X_d)^T$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, the probability density function is:*

$$\boxed{p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)} \quad (38)$$

The multidimensional Hermite polynomials $H_{\boldsymbol{\alpha}}(\mathbf{x})$ for multi-index $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)$ satisfy the orthogonality relation:

$$\boxed{\int_{\mathbb{R}^d} H_{\boldsymbol{\alpha}}(\mathbf{x})H_{\boldsymbol{\beta}}(\mathbf{x})p(\mathbf{x})d\mathbf{x} = \boldsymbol{\alpha}!\delta_{\boldsymbol{\alpha\beta}}} \quad (39)$$

where $\boldsymbol{\alpha}! = \alpha_1! \cdots \alpha_d!$ and $\delta_{\boldsymbol{\alpha\beta}}$ is the Kronecker delta.

Theorem A.2 (Isserlis-Wick Theorem for Gaussian Moments)**.** *For a centered multivariate Gaussian random vector* $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$*, the moments satisfy:*

$$\boxed{\mathbb{E}[X_{i_1} X_{i_2} \cdots X_{i_{2n}}] = \sum_{pairings} \prod_{pairs\ (j,k)} \Sigma_{i_j, i_k}} \tag{40}$$

where the sum is over all possible pairings of the $2n$ *indices into* n *pairs. For odd moments:*

$$\boxed{\mathbb{E}[X_{i_1} X_{i_2} \cdots X_{i_{2n+1}}] = 0} \tag{41}$$

The generating function for multidimensional Hermite polynomials is:

$$\boxed{\exp\left(\mathbf{t}^T \mathbf{x} - \frac{1}{2}\mathbf{t}^T \mathbf{\Sigma} \mathbf{t}\right) = \sum_{\boldsymbol{\alpha}} \frac{H_{\boldsymbol{\alpha}}(\mathbf{x})}{\boldsymbol{\alpha}!} \mathbf{t}^{\boldsymbol{\alpha}}} \tag{42}$$

where $\mathbf{t}^{\boldsymbol{\alpha}} = t_1^{\alpha_1} \cdots t_d^{\alpha_d}$.

Ongoing research: Numerical Implementation Strategy

From a numerical perspective, the key insight is to leverage these two fundamental properties—the orthogonality of multidimensional Hermite polynomials and the Isserlis-Wick theorem for Gaussian moments—to develop efficient computational strategies. Rather than hardcoding a Monte-Carlo integration specific formula for each individual moment calculation, we can implement robust quadrature rules that systematically evaluate these expectations. This approach transforms the computational burden from deriving explicit closed-form expressions for every moment to designing appropriate numerical integration schemes that exploit the underlying mathematical structure, making the analysis both more general and computationally tractable.

A.2 Personal contribution: Detailed Calculations for Two Hidden Layer MMNN

This section presents the complete development of calculations for two hidden layer MMNN (Matrix Memory Neural Networks), including exact derivations of the NTK and analysis of Fisher and Kibble distributions.

A.2.1 • Calculation of activation's correlations

Once we calculate the NTK of MMNN for 1 layer, we must solve the problem of calculating the expectation $\mathbb{E}[\mathrm{ReLU}(X_1)\mathrm{ReLU}(X_2)]$ for a pair of random variables (X_1, X_2) following a bivariate normal distribution. The approach consists of standardizing the variables and decomposing the integral into four terms corresponding to different moments of the standardized variables.

A.2.2 • Calculation of $\mathbf{I}_{11}$ (Probability)

The term $\mathbf{I}_{11}$ represents the probability $P(Z_1 > a, Z_2 > b)$ where Z_i are the standardized variables. It is calculated via the cumulative distribution function of the standard bivariate normal distribution, providing the foundational probability measure for our subsequent calculations.

A.2.3 • Special Case: Zero Means ($\mu_1 = \mu_2 = 0$)

In this case, the derivation leads to an elegant formula through Price's theorem, which was a central tool in our analysis. By integrating from 0 to ρ and adding the appropriate normalization constant, we obtain the final expression for the zero-mean case. This result forms the backbone of our theoretical understanding of ReLU interactions in the infinite-width limit.

$$\boxed{\mathbb{E}[\mathrm{ReLU}(Z_1)\mathrm{ReLU}(Z_2)] = \frac{1}{2\pi}(\sqrt{1-\rho^2} + \rho \arccos(-\rho))} \tag{43}$$

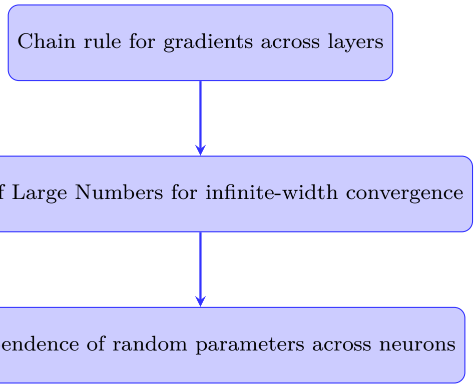
A.2.4 • Proof of the Recursion for the NTK

Now we can prove the recursion for the NTK.

Outline of Proof Structure. We establish this recursion by decomposing the total NTK at layer ℓ into contributions from different parameter groups:

Goal: We aim to show that the NTK at layer ℓ can be written as a sum of contributions from previous layers and the current layer.

Intuition: We use the chain rule to decompose gradients with respect to parameters at different layers, then apply the Law of Large Numbers in the infinite-width limit.



The total NTK at layer ℓ is the sum of contributions from all trainable parameters up to that layer:

$$\Theta^{(\ell)}_{rand}(x,x') = \underbrace{(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x))^\top (\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x'))}_{\text{Term 1}} + \underbrace{\sum_{j=1}^{n_\ell} \frac{\partial f^{(\ell)}(x)}{\partial A_j^{(\ell)}} \frac{\partial f^{(\ell)}(x')}{\partial A_j^{(\ell)}}}_{\text{Term 2}} + \underbrace{\frac{\partial f^{(\ell)}(x)}{\partial c^{(\ell)}} \frac{\partial f^{(\ell)}(x')}{\partial c^{(\ell)}}}_{\text{Term 3}} \tag{44}$$

Term 3 (Bias contribution): This term is deterministic and equals σ_c^2 from the bias parameter.

Term 2 (Layer ℓ weights contribution): The gradient with respect to a weight $A_j^{(\ell)}$ is:

$$\frac{\partial f^{(\ell)}(x)}{\partial A_j^{(\ell)}} = \frac{\sigma_A}{\sqrt{n_\ell}} \sigma(w_j^\top h^{(\ell-1)}(x) + \beta b_j) \tag{45}$$

Term 2 becomes:

$$\frac{\sigma_A^2}{n_\ell} \sum_{j=1}^{n_\ell} \sigma(w_j^\top h^{(\ell-1)}(x) + \beta b_j) \sigma(w_j^\top h^{(\ell-1)}(x') + \beta b_j) \tag{46}$$

For a fixed realization of the previous layer's output $h^{(\ell-1)}$, this is a sum of i.i.d. random variables over (w_j, b_j). By the Law of Large Numbers, as $n_\ell \to \infty$:

$$\text{Term 2} \xrightarrow{p} \sigma_A^2 \mathbb{E}_{w,b}\left[\sigma(\cdot)\sigma(\cdot)\right] = \sigma_A^2 \Sigma^{(\ell)}(x,x') \tag{47}$$

Term 1 (Previous layers contribution): Using the chain rule:

$$\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) = \frac{\partial f^{(\ell)}(x)}{\partial h^{(\ell-1)}(x)} \nabla_{\theta^{(\ell-1)}} h^{(\ell-1)}(x) \tag{48}$$

Term 1 becomes:

$$(\nabla_{\theta^{(\ell-1)}} h^{(\ell-1)}(x))^\top \left(\frac{\partial f^{(\ell)}}{\partial h^{(\ell-1)}}(x) \right)^\top \left(\frac{\partial f^{(\ell)}}{\partial h^{(\ell-1)}}(x') \right) (\nabla_{\theta^{(\ell-1)}} h^{(\ell-1)}(x')) \tag{49}$$

The first and last parts form the NTK of the previous layers, $\Theta^{(\ell-1)}(x,x')$. The middle part is a product of Jacobians that, as $n_\ell \to \infty$, converges to:

$$\left(\frac{\partial f^{(\ell)}}{\partial h^{(\ell-1)}} \right)^\top \left(\frac{\partial f^{(\ell)}}{\partial h^{(\ell-1)}} \right) \xrightarrow{p} \sigma_A^2 \dot{\Sigma}^{(\ell)}(x,x') \tag{50}$$

Thus:

$$\text{Term 1} \xrightarrow{p} \Theta^{(\ell-1)}(x,x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x,x') \tag{51}$$

Combining terms: In the infinite-width limit, we obtain:

$$\Theta^{(\ell)}(x,x') = \Theta^{(\ell-1)}(x,x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x,x') + \sigma_A^2 \Sigma^{(\ell)}(x,x') + \sigma_c^2 \tag{52}$$

Absorbing the σ_A^2 factor into the kernel definitions gives the stated recursion. □

Summary of Formulas

- **Layer Output:**

$$\boxed{h^{(\ell)}(x) = c^{(\ell)} + \frac{\sigma_A}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} h^{(\ell-1)}(x) + \beta b_j^{(\ell)}\right)}$$

- **NNGP Kernel Recursion:**
  - *Base Kernel:* Expectation over current layer's random parameters, conditioned on the previous layer's output GP $h^{(\ell-1)}$.

$$\boxed{\Sigma^{(\ell)}(x,x') = \mathbb{E}_{w,b}\left[\sigma\left(w^\top h^{(\ell-1)}(x) + \beta b\right) \sigma\left(w^\top h^{(\ell-1)}(x') + \beta b\right)\right]}$$

  - *Full Kernel:*

$$\boxed{\mathcal{K}_{\text{NNGP}}^{(\ell)}(x,x') = \sigma_A^2 \Sigma^{(\ell)}(x,x') + \sigma_c^2}$$

- **NTK Kernel Recursion:**
  - *Base Derivative Kernel:*

$$\boxed{\dot{\Sigma}^{(\ell)}(x,x') = \mathbb{E}_{w,b}\left[\dot{\sigma}\left(w^\top h^{(\ell-1)}(x) + \beta b\right) \dot{\sigma}\left(w^\top h^{(\ell-1)}(x') + \beta b\right) w^\top w\right]}$$

  - *Full Kernel:*

$$\boxed{\Theta^{(\ell)}(x,x') = \Theta^{(\ell-1)}(x,x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x,x') + \sigma_A^2 \Sigma^{(\ell)}(x,x') + \sigma_c^2}$$

A.3 Fisher and Kibble Distributions

Exact Probability Laws

The random variables $(\rho_1, \|x_1\|^2, \|y_1\|^2)$ follow exact distributions:
Fisher Distribution for correlation:

$$\boxed{\rho_1 \sim \text{Fisher}(\rho, r) = \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-\theta^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho\theta)^{r-\frac{3}{2}}} \times {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\theta}{2}\right)} \tag{53}$$

Kibble Distribution for squared norms:

$$\boxed{(\|x_1\|^2, \|y_1\|^2) \sim \text{Kibble}(r, \rho) = \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1}\rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right)} \tag{54}$$

B
DEEP NARROW NEURAL NETWORKS

B.1 Scaled NTK at Initialization

Theorem B.1 (Theorem 1: Scaled NTK at Initialization)**.** *For f_θ^L initialized appropriately, as $L \to \infty$:*

Deep Narrow NTK Limit

$$\boxed{\tilde{\Theta}_0^L(x, x') \xrightarrow{p} \tilde{\Theta}^\infty(x, x') = (x^T x' + 1 + \mathbb{E}_g[\sigma(g(x))\sigma(g(x'))])I_{d_{out}}} \tag{55}$$

where $g \sim GP(0, \rho^2 d_{in}^{-1} x^T x' + \beta^2)$ (Gaussian random field).

Alternative formulation: $\kappa_1(\cos(u) \cdot v)$ where:

- $v = \frac{1}{1+\beta^2/\alpha^2}$, where β is the bias variance
- $\alpha = \frac{\|x\|\|x'\|\rho}{d_{in}}$, where $\cos(u)$ is the cosine distance between x, x'

This framework provides a promising direction for analyzing deep narrow networks through limited expansion analysis.

B.2 Research Perspectives for Deep Narrow Networks

Architectural modifications:

- **Unit concatenation**: Combine multiple narrow networks to increase expressivity
- **Skip connections**: Investigate ResNet-style connections:

$$\tilde{\Theta}^\infty_{\text{skip}}(x, x') = \tilde{\Theta}^\infty(x, x') + \text{skip terms}$$

Initialization studies:

- **Alternative initializations**: Explore different schemes beyond current setup
- $\beta \to 0$ **limit**: Analyze behavior when bias variance vanishes:

$$v = \frac{1}{1 + \beta^2/\alpha^2} \to 1 \text{ as } \beta \to 0$$

- **Complex kernel structures**: Find initializations that yield more intricate kernel forms

Remark: Theoretical Extensions

Theoretical extensions:
- Extension of Hayou & Yang's work on ResNets showing "wide and deep limits commute"
- Mean field analysis for deep narrow frameworks
- Careful analysis of stochasticity in initialization (not dropout)
- Experimental validation on practical tasks